

LibroFlow – Library Management System

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Developed by:

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1. Project Overview

LibroFlow is an advanced web-based Library Management System designed to automate and modernize all library operations—book borrowing, returning, inventory tracking, and user management—within a single digital environment. The system supports multiple user roles (Admin, Department Manager, Supplier, Finance Manager, and User) through personalized dashboards and real-time data access. The main goal is to replace outdated manual systems with smart automation that enhances efficiency, minimizes human error, and increases user engagement. LibroFlow is scalable, adaptable for universities, public, or private libraries, and promotes reading habits through gamification and a point-based reward system.

2. Motivation and Problem Definition

Traditional library systems rely heavily on manual processes—recording transactions by hand, tracking inventory on spreadsheets, and managing suppliers through paper documentation. These methods lead to frequent human errors, lack of real-time visibility, poor user engagement, and inefficient financial management. LibroFlow solves these challenges through a unified digital system offering automation, transparency, and interaction.

3. System Objectives and Features

Objectives: • Automate all library operations (borrowing, returning, and archiving). • Provide real-time tracking for books, orders, and employees. • Enhance engagement via rewards and reading competitions. • Offer role-based dashboards for every stakeholder. • Support strategic decision-making through analytics and reports. Key Features: • Authentication and Role-Based Access. • Book Management and Archiving. • Reward System for timely returns and active reading. • Supplier and Order Management. • Finance Module for transactions and reports. • Admin Control with system-wide supervision.

4. Technical Architecture

Backend: C# with .NET Core and MS SQL Server — providing a stable and scalable foundation for database operations and APIs. Frontend: React JS, Tailwind CSS, HTML, and JavaScript — enabling a dynamic, responsive user interface with role-specific experiences. The system follows a multi-tier architecture: • Frontend Layer: Handles user interaction and role-based views. • Backend Layer: Manages logic, authentication, and data flow. • Database Layer: Stores book, user, and transaction records.

5. Methodology and Tools

LibroFlow follows the Agile methodology, emphasizing iterative development and continuous collaboration among team members. Development Tools: • Visual Studio Code, SQL Server Management Studio. • GitHub (version control), Notion (task tracking), Figma (UI design). • Communication: WhatsApp. Frameworks and Libraries: • React + TypeScript for the UI. • Tailwind CSS for styling. • .NET Core for backend logic. • MS SQL Server for data storage.

6. Expected Deliverables and Impact

Deliverables: • Fully functional web platform accessible from browsers. • Role-based dashboards for each user category. • Documentation and user manual for staff and students. • Database schema and integration with authentication. Impact: • Local: Enhances university library efficiency and student engagement. • Institutional: Reduces administrative workload and improves accuracy. • Educational: Encourages a culture of reading through digital incentives.

7. Conclusion

LibroFlow represents a forward-looking approach to academic library management. By integrating automation, analytics, and user engagement features within a clean, scalable web system, it sets a model for how modern libraries can function—efficient, data-driven, and student-centered. The use of C#/.NET Core and React.js ensures a robust backend with a responsive, accessible frontend—creating a cohesive digital ecosystem that supports both learning and management goals.